

Executive Summary

This manual is intended to serve as both a reference guide for the accompanying Macroeconomic Dashboard and an introduction to many of the primary economic data sources used by financial analysts, economists, and market participants. While the dashboard summarizes key economic indicators into a single, easy-to-interpret view, the purpose of this document is to explain where the data originates, how it is collected, when it is released, and how each indicator should be interpreted within the broader economic landscape.

Rather than relying on a single data provider, the dashboard intentionally utilizes information published by multiple official government agencies and recognized industry organizations, including the *Bureau of Economic Analysis* (BEA), *Bureau of Labor Statistics* (BLS), *Federal Reserve Economic Data* (FRED), *Federal Reserve Board*, *Institute for Supply Management* (ISM), and *ICE Data Indices*. Although many of these datasets can be accessed through multiple websites or aggregated into a single platform, this manual guides users to the original or most appropriate source whenever practical. The objective is not only to retrieve the required data but also to develop familiarity with the organizations responsible for producing it, understand how each report is constructed, and become comfortable navigating the resources commonly used throughout the financial industry.

Each section of this manual provides a brief overview of the reporting organization, publication schedule, methodology, and interpretation of the associated economic indicator before discussing how that indicator is incorporated into the dashboard. By understanding both the underlying data and the reasoning behind its inclusion, users are encouraged to move beyond simply recording economic statistics and begin evaluating how individual indicators contribute to a broader assessment of economic growth, labor market conditions, inflation, monetary policy, interest rates, and credit conditions.

Ultimately, the goal of this project is not simply to build a dashboard, but to develop a repeatable analytical process. By learning where economic data originates, how it is reported, and why each indicator matters, users can build a stronger foundation for interpreting macroeconomic conditions and making more informed, data-driven decisions.

Table of Contents

Each numbered section represents one of the dashboard's primary economic inputs.

Each section begins with a brief overview of the reporting organization and data source, followed by an explanation of how the indicator is collected, when it is published, how it should be interpreted, and the rationale for its inclusion in the dashboard.

Economic Growth

- 1.) [Gross Domestic Product \(GDP\)](#)
- 2.) [Institute for Supply Management \(ISM\)](#)
- 3.) [Industrial Production](#)

Labor Market

- 4.) [Nonfarm Payrolls](#)
- 5.) [Unemployment Rate](#)
- 6.) [Labor Force Participation Rate](#)
- 7.) [Wage Growth](#)

Inflation

- 8.) [Consumer Price Index/Core \(CPI\)](#)
- 9.) [Personal Consumption Expenditures/Core \(PCE\)](#)

Interest Rates

- 10.) [Effective Federal Funds Rate \(EFFR\)](#)
- 11.) [2-Year Treasury Constant Maturity Yield \(DGS2\)](#)
- 12.) [10-Year Treasury Constant Maturity Yield \(DGS10\)](#)
- 13.) [Treasury Yield Curve Spread \(T10Y2Y\)](#)

Credit Conditions

- 14.) [ICE BofA U.S. High Yield Index Option-Adjusted Spread \(BAMLH0A0HYM2\)](#)
- 15.) [ICE BofA U.S. Corporate Index Option-Adjusted Spread \(BAMLC0A0CM\)](#)

References

Economic Growth

The economic growth indicators used throughout this Dashboard are obtained from official U.S. government agencies and private industry organizations, including the ***Bureau of Economic Analysis*** (BEA), the ***Institute for Supply Management*** (ISM), the ***Federal Reserve Board***, and the ***Federal Reserve Economic Data*** (FRED) database. Together, these reports provide a comprehensive view of current economic activity across the United States.

Unlike Financial market indicators, which update daily, economic growth reports are generally released on a *monthly* or *quarterly* schedules and often undergo *revisions* as additional data becomes available. Consequently, economic growth should be evaluated by observing trends across multiple reporting periods rather than relying on any single release.

The indicators within this section measure different aspects of economic activity. Some evaluate overall economic output (***Gross Domestic Product***), while others assess *business activity*, *industrial production*, and the *pace of expansion or contraction* across key sectors of the economy. Because each report captures a different segment of economic activity, no single indicator should be interpreted in isolation.

Throughout this dashboard, economic growth indicators are used collectively to evaluate the current stage of the business cycle and determine whether economic activity is expanding, contracting, recovering, or slowing. When considered together, these measures provide a broad assessment of the strength and direction of the U.S. economy.

1.) GDP

- a. The ***Bureau of Economic Analysis*** (BEA) is a federal statistical agency that produces official U.S. economic data, including Gross Domestic Product (GDP). GDP measures the total value of all final goods and services produced within the United States and is the broadest measure of overall economic activity. GDP reports are released in three stages:
 - i. **Advance Estimate**- released about one month after the quarter ends
 - ii. **Second Estimate**- released a month later with more complete data
 - iii. **Third Estimate**- the final regular estimate, incorporating the most complete data available.

2.) ISM

- a. The ***Institute for Supply Management*** (ISM) is a nonprofit professional association that publishes monthly surveys measuring business conditions across the U.S economy. Its ***Purchasing Manager's Index*** (PMI) reports are widely regarded as leading economic indicators because they provide an early measure of economic activity before many government statistics are released.

- i. **Manufacturing (PMI)**- Measures business conditions in the U.S. manufacturing sector. A reading above 50 indicates expansion, while a reading below 50 indicates contraction. The Manufacturing PMI is a composite diffusion index, meaning it's calculated from five equally weighted subindexes.
 - 1. New Orders Index (20%)**
 - a. Measures incoming demand for manufactured goods.
 - 2. Production Index (20%)**
 - a. Measures current manufacturing output and production
 - 3. Employment Index (20%)**
 - a. Measures changes in manufacturing hiring and staffing
 - 4. Supplier Deliveries Index (20%)**
 - a. Measures supplier delivery times. This index is inverted, meaning slower deliveries contribute positively to the PMI because they typically reflect stronger demand and a busier economy.
 - 5. Inventories Index (20%)**
 - a. Measures changes in manufactures' inventory levels
- ii. **Services (PMI)**- Measures business conditions in the U.S. services sector. A reading above 50 indicates expansion, while a reading below 50 indicates contraction. The Services PMI is a composite diffusion index, meaning it's calculated from four equally weighted diffusion subindexes.
 - 1. Business Activity Index (25%)**
 - a. Measures the current level of business activity (Similar to production in manufacturing)
 - 2. New Orders Index (25%)**
 - a. Measures incoming demand for services.
 - 3. Employment Index (25%)**
 - a. Measures changes in hiring and staffing levels
 - 4. Supplier Deliveries Index (25%)**
 - a. Measures supplier delivery times. This index is inverted, meaning slower deliveries contribute positively to the PMI because they typically reflect stronger demand and a busier economy.

3.)Industrial Production

- a. The **Federal Reserve Board** publishes the ***Industrial Production and Capacity Utilization*** report each month, typically around the middle of the month. Industrial Production measures the real output of the U.S. industrial sector, including manufacturing, mining, and electric and gas utilities.
 - i. The index is reported relative to a base year of 2017 = 100 and is commonly analyzed on both a month-over-month (MoM) and year-over-year (YoY) basis to evaluate the direction and momentum of industrial activity.

Labor Market

The **Bureau of Labor Statistics** publishes the ***Employment Situation*** report each month, typically on the first Friday of each month at 8:30 a.m. Eastern Time. The report provides a comprehensive snapshot of the U.S. labor market and is based on two monthly surveys: the ***Current Employment Statistics*** (CES) survey and the ***Current Population Survey*** (CPS).

The ***Current Employment Statistics*** (CES) survey, commonly referred to as the ***Establishment Survey***, measures payroll employment, hours worked, and earnings from businesses and government agencies. The CES survey is the source of **Nonfarm Payrolls** and **Average Hourly Earnings**.

The ***Current Population Survey*** (CPS), commonly referred to as the ***Household Survey***, is used to measure unemployment, labor force participation, and other workforce characteristics. The CPS is the source of the **Unemployment Rate** and **Labor Force Participation Rate**.

(P) Indicates a preliminary estimate and is subject to revision in subsequent reports

4.) Nonfarm Payrolls

- a. Measures the monthly change in the number of paid employees in the U.S. economy, excluding farm workers, private household employees, active-duty military personnel, and the self-employed. It is one of the most closely watched indicators of labor market strength and overall economic activity. Nonfarm payrolls is reported as month-over-month change in employment (in thousands), not as the total number of jobs in the economy.
 - i. **Source:** *Current Employment Statistics (CES), Establishment Survey*

5.) Unemployment Rate

- a. Measures the percentage of the civilian labor force that is unemployed and actively seeking employment. Individuals who are not working and not actively looking for work are not included in the unemployment rate. A lower unemployment rate generally indicates a stronger labor market, while a higher unemployment rate may signal labor market weakness.

i. **Source:** *Current Population Survey, (CPS), Household Survey*

6.) Labor Force Participation

- a. Measures the percentage of the civilian non-institutional population that is either employed or actively seeking employment. An increasing labor force participation rate generally indicates more individuals are entering or remaining in the workforce, while a declining participation rate may suggest workers are leaving the labor force due to retirement, discouragement, education, or other factors.

i. **Source:** *Current Population Survey (CPS), Household Survey*

7.) Wage Growth

- a. **Average Hourly Earnings** (Wage Growth) measures the average hourly earnings of private-sector employees and is commonly used as an indicator of wage growth and labor cost inflation. Rising wage growth generally reflects a stronger labor market and increased earnings for workers but may also contribute to inflationary pressures. Slowing wage growth may indicate easing labor demand and reduced inflationary pressures.

i. **Source:** *Current Employment Statistics (CES), Establishment Survey*

Inflation

The **Bureau of Labor Statistics (BLS)** publishes the **Consumer Price Index (CPI)**, Typically around the second week of the month at 8:30 a.m. Eastern Time. The CPI report measures changes in prices paid by urban consumers for a market basket of consumer goods and services and is one of the primary measures of U.S. inflation.

The **Bureau of Economic Analysis (BEA)** publishes the **Personal Income and Outlays** report each month, typically near the end of the month at 8:30 a.m. Eastern Time. The report includes the **Personal Consumption Expenditures (PCE) Price Index**, which measures changes in the prices paid for goods and services consumed by households.

8.) Consumer Price Index (CPI) & Core (CPI)

- a. The report includes both Headline CPI (“All items”) and Core CPI (“All items less Food and Energy”) and reports inflation in both a month-over-month (MoM) and a 12-month (year-over-year, YoY) basis.
 - i. The CPI report contains both seasonally adjusted and not seasonally adjusted data
 - ii. Headline CPI is reported as “All Items”
 - iii. Core CPI is reported as "All Items Less Food and Energy”
 - iv. This Dashboard uses the 12-month (YoY) inflation rate to evaluate long-term inflation trends and compare inflation against the FED 2% inflation objective.

9.) Personal Consumption Expenditures (PCE) & Core (PCE)

- a. The report includes both **Headline PCE** and **Core PCE** (*Excluding Food and Energy*) and reports inflation on both a month-over-month (MoM) and 12-month (year-over-year, YoY) basis.
 - i. Headline PCE includes all goods and services
 - ii. Core PCE excludes Food and Energy
 - iii. The Federal Reserve’s 2% inflation target is measured using the PCE Price Index, although both CPI and PCE are monitored.
 - iv. This Dashboard uses the 12-month (YoY) inflation rate to evaluate long-term inflation trends and compare inflation against the FED 2% inflation objective.

Interest Rates

The ***Federal Reserve Economic Data*** (FRED) database, maintained by the ***Federal Reserve Bank of St. Louis***, serves as the primary source for the interest rate data used throughout this dashboard. FRED aggregates economic and financial data from numerous official organizations, including the ***Federal Reserve Board***, the ***U.S. Treasury***, the ***Bureau of Labor Statistics*** (BLS), the ***Bureau of Economic Analysis*** (BEA), and other government agencies, into a centralized database.

Most interest rate series are published on a daily basis, although many are also available as weekly, monthly, quarterly, or annual averages. Unless otherwise noted, this dashboard uses most recent daily observations to provide a current snapshot of the financial market conditions rather than historical averages.

When using FRED, users should understand that many market interest rates are reported as benchmark series rather than the yields of individual securities. For example, Treasury “Constant Maturity” rates are estimated from the U.S. Treasury yield curve using market quotations from actively traded Treasury Securities, providing standardized yields at fixed maturities such as 2 and 10 years.

10.) Federal Funds Effective Rate

- a. The ***Effective Federal Funds Rate*** (EFFR) represents the volume-weighted median interest rate at which the U.S. depository institutions lend reserve balances to one another on an overnight basis. It is the Federal Reserve’s Primary operating target for implementing monetary policy and serves as the benchmark for short-term interest rates throughout the U.S. financial system.
 - i. The EFFR reflects the actual overnight borrowing costs between banks rather than the Federal Reserve’s target range.
 - ii. Changes in the federal funds rate influence borrowing costs across the economy, including consumer loans, mortgages, business financing, and other short-term interest rates.
 - iii. The ***Federal Open Market Committee*** (FOMC) establishes a target range for the federal funds rate at its scheduled policy meetings (typically eight per year). The Federal Reserve then conducts monetary policy operations to keep the effective rate trading within that target range.
 - iv. The ***Federal Reserve Bank of New York*** calculates and publishes the Effective Federal Funds Rate each business day using transaction-level data from overnight federal funds trades. FRED updates the series after the official release.
 - v. This dashboard uses the daily Effective Federal Funds Rate (EFFR) to classify monetary policy as Accommodative, Neutral, or Restrictive, providing a current assessment of the Federal Reserve’s policy stance.

11.) 2-Year Treasury Constant Maturity Yield (DGS2)

- a. The 2-Year Treasury Constant Maturity Yield (DGS2) represents the estimated yield on a hypothetical U.S. Treasury security with exactly two years remaining until maturity. Rather than reflecting the yield of a single Treasury note, the rate is interpolated from the U.S. Treasury yield curve using market quotations from actively traded Treasury Securities.

- i. The 2-Year Treasury is considered one of the best indicators of market expectations for future Federal Reserve monetary policy and short-term interest rates.
- ii. Because its maturity is relatively short, the 2-year Treasury yield is highly sensitive to anticipated changes in the Federal Funds Rate and often moves in advance of Federal Reserve policy decisions.
- iii. The U.S. Treasury constructs the Constant Maturity series each business day by fitting a yield curve to market quotations from actively traded treasury securities and interpolating yields at standardized maturities. FRED publishes the resulting benchmark series.
- iv. As a U.S. Treasury security, the 2-year Treasury is considered effectively free of default risk and serves as a benchmark for pricing a wide range of short- and intermediate-term financial assets.
- v. This dashboard uses the daily 2-year Treasury Constant Maturity Yield (DGS2) to evaluate short-term interest rate expectations and as one component in calculating the Treasury Yield Curve Spread.

12.) 10-Year Treasury Constant Maturity Yield (DGS10)

- a. The ***10-Year Treasury Constant Maturity Yield*** (DGS10) represents the estimated yield on a hypothetical U.S. Treasury security with exactly ten years remaining until maturity. Like the 2-Year Treasury, it is an interpolated benchmark rate derived from the U.S. Treasury yield curve using market quotations from actively traded Treasury Securities rather than the yield of a single Treasury note.
 - i. The 10-Year Treasury is widely regarded as the benchmark for long-term interest rates in the United States and serves as a reference rate for mortgages, corporate bonds, and other long-term borrowing.
 - ii. The 10-Year Treasury reflects market expectations for long-term economic growth, inflation, and future monetary policy. Unlike the 2-Year Treasury, it is influenced more heavily by long-term economic fundamentals than by near-term Federal Reserve policy expectations.
 - iii. The U.S. Treasury constructs the Constant Maturity series each business day by fitting a yield curve to market quotations from actively traded treasury securities and interpolating yields at standardized maturities. FRED publishes the resulting benchmark series.
 - iv. As a U.S. Treasury security, the 10-Year Treasury note is considered effectively free of default risk and serves as one of the most important benchmark interest rates in global financial markets.

- v. This dashboard uses the daily 10-Year Treasury Constant Maturity Yield (DGS10) to evaluate long-term market expectations and as one component in calculating the Treasury Curve Spread.

13.) 10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity (T10Y2Y)

- a. The *Yield Curve Spread* represents the difference between the *10-Year Treasury Constant Maturity Yield* (DGS10) and the *2-Year Treasury Constant Maturity Yield* (DGS2). The *spread* is one of the most widely followed indicators of market expectations for future economic growth, inflation, and monetary policy. It is commonly used by economists and financial market participants to assess the shape of the U.S. Treasury yield curve.
 - i. A **Positive (normal)** yield curve occurs when the long-term Treasury yields exceed short-term Treasury yields, reflecting expectations for continued economic growth and normal market conditions.
 - ii. A **Flat** yield curve indicates that long-term and short-term Treasury yields are similar, often suggesting increased economic uncertainty or a transition between monetary policy cycles.
 - iii. An **Inverted** yield curve occurs when short-term Treasury yields exceed long-term Treasury yields. Historically, sustained yield curve inversions have frequently preceded U.S. economic recessions, although they should not be interpreted as a guarantee that a recession will occur.
 - iv. Although FRED publishes the *10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity* (T10Y2Y) as a standalone series, this dashboard calculates the spread directly by subtracting the displayed *2-Year Constant Maturity Yield* (DGS2) from the *10-Year Treasury Constant Maturity Yield* (DGS10). This approach ensures consistency between the reported Treasury yields and calculated spread.
 - v. Because the published Treasury yields are rounded to two decimal places, the calculated spread may *occasionally* differ from the official FRED T10Y2Y series by **one basis point** (0.01%) due to rounding. This difference is immaterial and does not affect the dashboard's interpretation of market conditions.
 - vi. This Dashboard classifies the Yield Curve as **Normal**, **Flat**, and **Inverted** to provide a concise assessment of market expectations and the relationship between short-term and long-term interest rates.

Credit Conditions

The **Federal Reserve Economic Data** (FRED) database serves as the primary source for the credit market indicators used throughout this dashboard. The credit spread data is produced by ICE Data Indices, LLC (formerly ICE BofA) and made available through FRED as daily market-based measures of corporate credit risk.

Unlike Treasury yields, which represent the interest rates paid on U.S. government debt, corporate credit spreads measure the additional yield investors require to hold corporate bonds instead of comparable U.S. Treasury securities. This additional yield, commonly referred to as the credit spread or **Risk Premium**, compensates investors for the possibility of default and other credit-related risks.

Corporate bond spreads are calculated as Option-Adjusted Spreads (OAS), which measure the difference between a corporate bond's yield and a comparable Treasury Yield after adjusting for embedded options such as issuer call provisions. As a result, OAS provides a more accurate measure of the markets assessment of credit risk than a simple yield curve spread.

Credit spread data is updated each business day using current market prices and yields from the corporate bond market. Unless otherwise noted, this dashboard uses the **most recent daily observation** to provide a current assessment of credit market conditions.

In general, narrow (tight) credit spreads indicate that investors perceive relatively low credit risk and are willing to accept lower compensation for holding corporate bonds, reflecting favorable financial conditions. Conversely, wide credit spreads indicate increased concern regarding economic conditions, corporate defaults, or overall market risk, causing investors to demand higher risk premiums.

14.) ICE BofA US High Yield Index Option-Adjusted Spread (BAMLH0A0HYM2)

- a. The **ICE BofA US High Yield Index Option-Adjusted Spread** (OAS) measures the average yield premium investors require to hold below-investment grade (“junk”) corporate bonds instead of comparable U.S. Treasury securities. Because high-yield issuers generally carry greater credit risk, this spread serves as an important indicator of investor risk appetite and perceived default risk.
 - i. High-Yield bonds are corporate bonds rated *below* **BBB** – (S&P/Fitch) or *below* **Baa3** (Moody's) and are considered speculative-grade securities.
 - ii. A **Narrow** (tight) high yield spread indicates that investors are comfortable accepting relatively low compensation for credit risk, reflecting favorable financial conditions and strong market confidence.

- iii. A **Wide** high yield spread indicates that investors are demanding higher compensation for assuming credit risk, often reflecting increased economic uncertainty or concerns regarding future corporate defaults
- iv. The ICE BofA High Yield Option-Adjusted Spread is calculated each business day using market prices of eligible high yield corporate bonds after adjusting for embedded options such as issuer call provisions. FRED publishes the resulting benchmark series.
- v. This Dashboard uses the daily High Yield Option-Adjusted Spread to evaluate investor risk appetite and overall conditions within the speculative-grade corporate bond market.

15.) ICE BofA U.S. Corporate Index Option-Adjusted Spread (BAMLC0A0CM)

- a. The *ICE BofA U.S. Corporate Index Option-Adjusted Spread* (OAS) measures the average yield premium investors require to hold investment-grade corporate bonds instead of comparable U.S. Treasury securities. Because investment-grade issuers generally possess stronger credit quality, these spreads are typically lower and more stable than High Yield spreads.
 - i. Investment-grade corporate bonds are generally rated **BBB** – (S&P/Fitch) or *higher* or **Baa3** (Moody’s) or *higher*, indicating relatively strong credit quality and lower expected default risk.
 - ii. A **Narrow** (tight) investment-grade spread reflects favorable credit conditions and strong investor confidence in the financial health of high-quality corporate issuers.
 - iii. A **Wide** investment-grade spread indicates increasing concern regarding corporate credit quality or broader economic conditions, causing investors to require greater compensation for credit risk.
 - iv. The *ICE BofA Corporate Option-Adjusted Spread* is calculated each business day using market prices of eligible investment-grade corporate bonds after adjusting for embedded options. FRED publishes the resulting benchmark series.
 - v. This Dashboard uses the daily *Investment Grade Option-Adjusted Spread* to evaluate overall corporate credit conditions and the availability of financing for higher-quality borrowers.

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